

TEMT freight emissions report

Illustrative India consumer group scenario

Illustrative scenario. SYNTHETIC DATA.

2025-10-01 to 2026-09-30 | Mode: all | Unit: all

Emissions 453.64 tCO2e	Shipments 24	Freight activity 1,13,23,897 tkm
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Emissions by mode

Road		268.58 tCO2e
Rail		27.71 tCO2e
Air		105.23 tCO2e
Ocean		52.11 tCO2e

Method and provenance

Emissions kgCO2e = tonnes x kilometres x profile factor. Engine: temt-demo-1.0.0.

Calculated: 2026-09-13T20:09:04.257Z. Processing: local.

Published factors are illustrative. This report uses a separate demonstration engine. It is not covered by the SGS validation of the TEMT v1.3 methodology against ISO 14083:2023.

Road HCV, India rail, container ocean at 10t/TEU, and distance-specific air profiles are used.

Selected filters are applied after validation. A shipment with an invalid leg is excluded in full.

Validation: 0 exception records; 0 excluded rows; 0 identifiable excluded shipments from 32 input rows. Counts cover the full input before filters.

Full factor values, assumptions and source references are included in the methodology appendix.

Shipment ledger

Shipment / leg	Date	Mode	Tonnes	km	Tonne-km	kgCO2e
FMC-1001 / 1	2025-10-07	road	185.6	1,420	2,63,552	17,473.50
FMC-1001 / 2	2025-10-07	road	185.6	38	7,053	467.60
FMC-1002 / 1	2025-11-10	road	409.6	362	1,48,275	9,830.65
FMC-1003 / 1	2025-12-13	road	249.6	684	1,70,726	11,319.16
FMC-1004 / 1	2026-01-16	rail	473.6	1,566	7,41,658	7,861.57
FMC-1004 / 2	2026-01-16	road	473.6	47	22,259	1,475.78
FMC-1005 / 1	2026-02-19	road	313.6	710	2,22,656	14,762.09
FMC-1006 / 1	2026-03-22	air	43.0	1,672	71,896	56,654.05
FMC-1007 / 1	2026-04-07	ocean	641.9	2,194	14,08,329	16,125.36
FMC-1007 / 2	2026-04-07	road	641.9	56	35,946	2,383.25
FMC-1008 / 1	2026-05-10	road	217.6	876	1,90,618	12,637.95
FMC-1009 / 1	2026-06-13	rail	441.6	1,420	6,27,072	6,646.96
FMC-1010 / 1	2026-07-16	road	281.6	362	1,01,939	6,758.57
FMC-1010 / 2	2026-07-16	road	281.6	65	18,304	1,213.56
FMC-1011 / 1	2026-08-19	road	505.6	684	3,45,830	22,928.56
FMC-1012 / 1	2026-09-22	road	345.6	1,566	5,41,210	35,882.20
FMC-1013 / 1	2025-10-07	road	185.6	710	1,31,776	8,736.75
FMC-1013 / 2	2025-10-07	road	185.6	74	13,734	910.59
FMC-1014 / 1	2025-11-10	rail	409.6	1,672	6,84,851	7,259.42
FMC-1015 / 1	2025-12-13	ocean	424.3	2,346	9,95,408	11,397.42
FMC-1016 / 1	2026-01-16	road	473.6	876	4,14,874	27,506.12
FMC-1016 / 2	2026-01-16	road	473.6	83	39,309	2,606.17
FMC-1017 / 1	2026-02-19	air	25.1	1,420	35,642	48,580.05
FMC-1018 / 1	2026-03-22	road	537.6	362	1,94,611	12,902.72
FMC-1019 / 1	2026-04-07	rail	377.6	684	2,58,278	2,737.75
FMC-1019 / 2	2026-04-07	road	377.6	92	34,739	2,303.21
FMC-1020 / 1	2026-05-10	road	217.6	1,566	3,40,762	22,592.49
FMC-1021 / 1	2026-06-13	road	441.6	710	3,13,536	20,787.44
FMC-1022 / 1	2026-07-16	road	281.6	1,672	4,70,835	31,216.37
FMC-1022 / 2	2026-07-16	road	281.6	101	28,442	1,885.68
FMC-1023 / 1	2026-08-19	ocean	859.5	2,498	21,47,031	24,583.50
FMC-1024 / 1	2026-09-22	rail	345.6	876	3,02,746	3,209.10

Methodology / factor register

Published well-to-wheel default factors. These are estimates, not measured carrier-specific factors. This demonstration is separate from production TEMT; values are applied without intermediate rounding.

road-hcv

India diesel HCV, 30-50 t GVW

0.0663 kgCO₂e / tonne-km | GLEC v3.2 (October 2025) | Source 1, page 106

- Indian heavy commercial vehicle category 3; 20-40 t payload capacity.
- Average load 68.3%; empty running 14.4%.
- The published factor already includes a 5% distance adjustment.
- Indian default factors are preliminary and subject to review.

rail-india

India rail, mixed traction

0.0106 kgCO₂e / tonne-km | GLEC v3.2 (October 2025) | Source 1, page 97

- Indian average combining diesel and electric traction.
- A national default, not a measured factor for a particular train or corridor.
- Indian default factors are preliminary and subject to review.

ocean-container

Dry container, Middle East / India

0.01145 kgCO₂e / tonne-km | GLEC v3.2 (October 2025) | Source 1, page 112

- Intra Middle East / India dry-container end-user factor: 114.5 g CO₂e per TEU-km.
- Converted using an illustrative average of 10 tonnes per TEU.
- The published factor includes 70% fleet load and a 15% distance adjustment.
- Port operations and first/last-mile legs are excluded unless supplied separately.

air-short

Air, short-haul mixed aircraft

1.363 kgCO₂e / tonne-km | GLEC v3.2 (October 2025) | Source 1, page 94

- Unknown aircraft mix: 55% belly freight, 45% freighter.
- Short-haul profile. This demo includes the 1,500 km boundary in short-haul.
- The published factor already includes the 95 km distance adjustment.
- Default emissions only; no separate aviation radiative-forcing multiplier.

air-long

Air, long-haul mixed aircraft

0.788 kgCO₂e / tonne-km | GLEC v3.2 (October 2025) | Source 1, page 94

- Unknown aircraft mix: 55% belly freight, 45% freighter.
- Long-haul profile for a leg longer than 1,500 km.
- The published factor already includes the 95 km distance adjustment.
- Default emissions only; no separate aviation radiative-forcing multiplier.

Source 1 / published factors

Smart Freight Centre, GLEC Framework v3.2

https://d1c2gz5q23tkk0.cloudfront.net/shrine_store/uploads/networks/3660/communication_news/13311209/61c644c3cbc7aabd60d91cf8d37cff2.pdf

Mode-shift illustration: keep road tonne-kilometres constant, then apply the selected rail share using the rail default. Savings are the road baseline less the blended road/rail scenario. Rail availability, capacity, detours, lead times and cost are not modelled.